

internal_services

Internal services + infrastructure

Inventory of every feature under `helix_code/internal/*` (72 packages). Assessed from code evidence (impl reality, wiring, `_test.go` presence) per CONST-035 / §11.4.107 anti-bluff. 📺 Video is no for every row (recordings are the conductor's job once a real analyzed recording exists); Overall is never confirmed for the same reason. Origin is native (HelixCode's own code).

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	📺
service	internal/adapters	il8n message resolution	done	yes	yes	unit	no	n
infrastructure	internal/adapters	translator injection seam	done	yes	yes	unit	no	n
infrastructure	internal/adapters	speckit debate adapter	partial	yes	unknown	unit	no	n
infrastructure	internal/adapters	container adapter	partial	yes	unknown	unit	no	n
service	internal/agent	agent execution interface	done	yes	yes	unit	no	n
service	internal/agent	agent capability management	done	yes	yes	unit	no	n
service	internal/agent	agent health check	done	yes	yes	unit	no	n
service	internal/agent	agent collaboration	done	yes	yes	unit	no	n
infrastructure	internal/agentbridge	verifier bridge	partial	yes	unknown	unit	no	n
service	internal/approval	approval request management	done	yes	yes	unit	no	n
service	internal/approval	approval mode selector	done	yes	yes	unit	no	n
infrastructure	internal/approvalwire	yes/no prompter	done	yes	yes	unit	no	n
service	internal/auth		done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
		user authentication						
service	internal/auth	session management	done	yes	yes	unit	no	n
service	internal/auth	password hashing (bcrypt/argon2)	done	yes	yes	unit	no	n
service	internal/auth	JWT token generation	done	yes	yes	unit	no	n
infrastructure	internal/autocommit	git auto-commit	done	yes	yes	unit	no	n
infrastructure	internal/autocommit	secret filter	done	yes	yes	unit	no	n
infrastructure	internal/cache	multi-tier cache	done	yes	yes	unit	no	n
infrastructure	internal/cache	redis tier	done	yes	yes	unit	no	n
infrastructure	internal/cache	disk tier	done	yes	yes	unit	no	n
infrastructure	internal/checkpoint	checkpoint manager	partial	yes	unknown	unit	no	n
service	internal/clarification	clarification engine	done	yes	yes	unit	no	n
service	internal/clarification	question generation	done	yes	yes	unit	no	n
infrastructure	internal/clientcore	agentic-tools provider	partial	yes	unknown	unit	no	n
infrastructure	internal/clientcore	skills provider	partial	yes	unknown	unit	no	n
service	internal/cognee	cognee client	done	yes	yes	unit	no	n
service	internal/cognee	cognee manager	done	yes	yes	unit	no	n
infrastructure	internal/cognee	cognee cache manager	done	yes	yes	unit	no	n
service	internal/commands	aider command	done	yes	yes	unit	no	n
service	internal/commands	approval command	done	yes	yes	unit	no	n
service	internal/commands	browser command	done	yes	yes	unit	no	n
service	internal/commands	edit command	done	yes	yes	unit	no	n
service			done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
		git auto-commit command						
service	internal/commands	mcp command	done	yes	yes	unit	no	no
service	internal/commands	markdown commands	done	yes	yes	unit	no	no
service	internal/commands	skills command	done	yes	yes	unit	no	no
service	internal/commands	subagents command	done	yes	yes	unit	no	no
service	internal/commands	tasks command	done	yes	yes	unit	no	no
service	internal/commands	worktree command	done	yes	yes	unit	no	no
infrastructure	internal/commands	command executor	done	yes	yes	unit	no	no
infrastructure	internal/commands	command registry	done	yes	yes	unit	no	no
infrastructure	internal/commands	command parser	done	yes	yes	unit	no	no
infrastructure	internal/config	viper-based config loader	done	yes	yes	unit	no	no
infrastructure	internal/config	cognee configuration management	done	yes	yes	unit	no	no
infrastructure	internal/config	verifier configuration	done	yes	yes	unit	no	no
infrastructure	internal/config	platform-UI adapters config	partial	yes	unknown	unit	no	no
infrastructure	internal/context	context builder	done	yes	yes	unit	no	no
infrastructure	internal/context	token counter	done	yes	yes	unit	no	no
infrastructure	internal/context	history condenser	done	yes	yes	unit	no	no
service	internal/continua	completion engine	done	yes	yes	unit	no	no
service	internal/continua	continue-edit tool	done	yes	yes	unit	no	no
service	internal/continua	continue-complete tool	done	yes	yes	unit	no	no
infrastructure	internal/database	database connection pool	done	yes	yes	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
infrastructure	internal/database	postgres integration	done	yes	yes	integ	no	n
infrastructure	internal/deployment	production deployer	done	yes	yes	unit	no	n
infrastructure	internal/deployment	deployment strategy	done	yes	yes	unit	no	n
service	internal/discovery	service registry with TTL + health checks	done	yes	yes	unit	no	n
service	internal/discovery	UDP multicast broadcast discovery	done	yes	unknown	unit	no	n
service	internal/discovery	dynamic port allocation	done	yes	yes	unit	no	n
service	internal/discovery	health monitoring (HTTP/gRPC/TCP)	done	yes	yes	unit	no	n
service	internal/editor	unified-diff code editing	done	yes	yes	unit	no	n
service	internal/editor	whole-file replacement editing	done	yes	yes	unit	no	n
service	internal/editor	search-and-replace editing	done	yes	yes	unit	no	n
service	internal/editor	line-range editing	done	yes	yes	unit	no	n
service	internal/editor	model-specific format selection	done	yes	yes	unit	no	n
service	internal/ensembleui	ensemble UI rendering	partial	partial	unknown	none	no	n
service	internal/event	publish-subscribe event bus	done	yes	yes	unit	no	n
service	internal/event	async + sync event handling	done	yes	yes	unit	no	n
service	internal/event	task/workflow/worker event types	done	yes	yes	unit	no	n
service	internal/fix	security issue auto-fix	done	yes	unknown	unit	no	n
service	internal/focus	hierarchical focus management	done	yes	unknown	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
service	internal/focus	priority-based focus tracking	done	yes	unknown	unit	no	no
service	internal/focus	focus chain tracking	done	yes	unknown	unit	no	no
service	internal/hardware	CPU detection + profiling	done	yes	yes	unit	no	no
service	internal/hardware	GPU detection (NVIDIA/AMD/Apple/Intel)	done	yes	yes	unit	no	no
service	internal/hardware	optimal LLM model-size inference	done	yes	yes	unit	no	no
service	internal/helixqa	HelixQA test wrapper	partial	yes	unknown	unit,integ	no	no
service	internal/hooks	hook registration + triggering	done	yes	yes	unit,integ	no	no
service	internal/hooks	priority-based hook execution	done	yes	yes	unit	no	no
service	internal/hooks	async/sync hook execution	done	yes	yes	unit	no	no
infrastructure	internal/i18nwiring	i18n catalog wire-all (multi-lang)	done	yes	yes	unit	no	no
infrastructure	internal/infraboot	on-demand infra boot (EnsureInfra)	done	yes	unknown	unit,integ	no	no
service	internal/kilocode	call-graph build + query	done	yes	unknown	unit	no	no
service	internal/kilocode	symbol rename engine	done	yes	unknown	unit	no	no
service	internal/kilocode	change-impact analyzer	done	yes	unknown	unit	no	no
service	internal/kilocode	refactor (extract-method/inline-call)	done	yes	unknown	unit	no	no
service	internal/kilocode	multi-edit tool	done	yes	unknown	unit	no	no
service	internal/llm	OpenAI provider (GPT models)	done	yes	yes	unit,integ	no	no
service	internal/llm	Anthropic provider	done	yes	yes	unit,integ	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
		(Claude models)						
service	internal/llm	Google Gemini provider	done	yes	yes	unit	no	n
service	internal/llm	Azure OpenAI provider	done	yes	yes	unit	no	n
service	internal/llm	AWS Bedrock provider	done	yes	yes	unit	no	n
service	internal/llm	Ollama local provider	done	yes	yes	unit,integ	no	n
service	internal/llm	llama.cpp local inference	done	yes	yes	unit	no	n
service	internal/llm	Mistral provider	done	yes	yes	unit	no	n
service	internal/llm	DeepSeek provider	done	yes	yes	unit	no	n
service	internal/llm	Groq provider	done	yes	yes	unit	no	n
service	internal/llm	streaming response handling	done	yes	yes	unit,integ	no	n
service	internal/llm	token counting + accounting	done	yes	yes	unit	no	n
service	internal/llm	model discovery/ listing	done	yes	yes	unit	no	n
service	internal/llm	load balancing across providers	done	yes	yes	unit	no	n
service	internal/llm	provider health monitoring	done	yes	yes	unit	no	n
service	internal/llm	ensemble provider orchestration	done	yes	yes	unit	no	n
service	internal/llm	embeddings generation	done	yes	yes	unit	no	n
infrastructure	internal/logging	structured logging with levels	done	yes	yes	unit	no	n
infrastructure	internal/logging	named loggers	done	yes	yes	unit	no	n
infrastructure	internal/logo	logo processing + assets	done	yes	unknown	unit	no	n
infrastructure	internal/logo		done	yes	unknown	none	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
		color extraction + icon generation						
infrastructure	internal/mcp	Model Context Protocol server	done	yes	partial	unit,integ	no	n
infrastructure	internal/mcp	JSON-RPC 2.0 tool invocation	done	yes	yes	unit	no	n
infrastructure	internal/mcp	WebSocket session management	done	yes	yes	unit	no	n
infrastructure	internal/mcp	OAuth token management	done	yes	unknown	unit	no	n
service	internal/memory	Cognee LLM memory integration	done	yes	unknown	unit,integ	no	n
infrastructure	internal/memory	memory state persistence	done	yes	unknown	unit	no	n
infrastructure	internal/monitoring	system metrics collection	done	yes	unknown	unit	no	n
infrastructure	internal/monitoring	health check monitoring	partial	partial	unknown	unit	no	n
service	internal/notification	multi-channel notification engine	done	yes	unknown	unit	no	n
service	internal/notification	Slack integration	done	yes	unknown	unit	no	n
service	internal/notification	Discord integration	done	yes	unknown	unit	no	n
service	internal/notification	Telegram integration	done	yes	unknown	unit	no	n
service	internal/notification	Email/SMTP integration	done	yes	unknown	unit	no	n
service	internal/notification	Teams integration	done	yes	unknown	unit	no	n
service	internal/notification	rate limiting + retry	done	yes	unknown	unit	no	n
infrastructure	internal/performance	performance optimizer	done	yes	unknown	unit	no	n
infrastructure	internal/performance	CPU optimization	partial	partial	unknown	unit	no	n
infrastructure	internal/performance	memory optimization	partial	partial	unknown	unit	no	n
infrastructure	internal/persistence	file-based state store	done	yes	unknown	unit	no	n
infrastructure			done	yes	unknown	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Impact
	internal/	session						
	internal/	serialization						
infrastructure	persistence							
	internal/	auto-save	done	yes	unknown	unit	no	n
	internal/	manager						
service	persistence							
	internal/	task executor	done	yes	unknown	unit	no	n
	internal/							
service	planner	OpenHands	partial	partial	unknown	unit	no	n
	internal/	plan execution						
	internal/							
infrastructure	planner							
	internal/	plan tree	done	yes	unknown	unit	no	n
	internal/	system						
infrastructure	plantree							
	internal/	plan node	done	yes	unknown	unit	no	n
	internal/	management						
infrastructure	plantree							
	internal/	plan branching	done	yes	unknown	unit	no	n
	internal/	+ merging						
service	plantree							
	internal/	plugin base	done	yes	unknown	unit	no	n
	internal/	framework						
service	plugins							
	internal/	plugin	done	yes	unknown	unit	no	n
	internal/	activation						
service	plugins							
	internal/	plugin hooks	partial	partial	unknown	unit	no	n
	internal/	registry						
service	plugins							
	internal/project	project	done	yes	unknown	unit	no	n
	internal/project	manager						
service								
	internal/project	project	done	yes	unknown	unit	no	n
	internal/project	metadata						
service								
	internal/project	database-	done	yes	unknown	unit	no	n
	internal/project	backed storage						
infrastructure								
	internal/	memory loader	done	yes	unknown	unit	no	n
	internal/							
infrastructure	projectmemory							
	internal/	filesystem	done	yes	unknown	unit	no	n
	internal/	watcher						
infrastructure	projectmemory							
	internal/	memory	done	yes	unknown	unit	no	n
	internal/	registry						
infrastructure	projectmemory							
	internal/	provider	done	yes	unknown	unit	no	n
	internal/	interface						
infrastructure	provider							
	internal/	multi-provider	done	yes	unknown	unit	no	n
	internal/	support						
service	provider							
	internal/	AI integration	done	yes	unknown	unit,integ	no	n
	internal/							
service	providers							
	internal/	fallback + load	done	yes	unknown	unit,integ	no	n
	internal/	balancing						
service	providers							
	internal/	quality gate	done	yes	unknown	unit	no	n
	internal/	scoring						
infrastructure	quality							
	internal/	build	done	yes	unknown	unit	no	n
	internal/	verification						
infrastructure	quality							

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
infrastructure	internal/quality	linting validation	done	yes	unknown	unit	no	no
infrastructure	internal/redis	redis client	done	yes	unknown	unit	no	no
infrastructure	internal/redis	key-value operations	done	yes	unknown	unit	no	no
infrastructure	internal/redis	pub/sub messaging	done	yes	unknown	unit	no	no
service	internal/render	ANSI renderer	done	yes	unknown	unit	no	no
service	internal/render	fancy-mode output	done	yes	unknown	unit	no	no
service	internal/render	frame buffer management	done	yes	unknown	unit	no	no
service	internal/render	streaming block renderer	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	symbol extraction	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	file ranking engine	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	repo cache	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	tree-sitter integration	done	yes	unknown	unit	no	no
service	internal/roocode	Roo-Code CLI port	partial	partial	unknown	unit	no	no
service	internal/roocode	template-based code generation	partial	partial	unknown	unit	no	no
service	internal/rules	rule hierarchy management	done	yes	yes	unit	no	no
service	internal/rules	rule pattern matching (glob/regex/exact)	done	yes	yes	unit	no	no
service	internal/rules	rule category + tag querying	done	yes	yes	unit	no	no
service	internal/secrets	secret loader from .env files	done	yes	yes	unit	no	no
service	internal/secrets	secret validation + missing-var detection	done	yes	yes	unit	no	no
service	internal/security		done	yes	yes	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
		security manager (global + local)						
service	internal/security	feature scanning + zero-tolerance validation	done	yes	yes	unit	no	n
service	internal/server	auth endpoints (register/login/logout/refresh)	done	yes	unknown	unit	no	n
service	internal/server	user profile endpoints (GET/PUT / me)	done	yes	unknown	unit	no	n
service	internal/server	worker mgmt endpoints (list/register/heartbeat)	done	yes	unknown	unit	no	n
service	internal/server	task CRUD endpoints	done	yes	unknown	unit	no	n
service	internal/server	project CRUD endpoints	done	yes	unknown	unit	no	n
service	internal/server	workflow execution endpoints	done	yes	unknown	unit	no	n
service	internal/server	system stats + health-check endpoints	done	yes	unknown	unit	no	n
service	internal/server	LLM provider + model list endpoints	done	yes	unknown	unit	no	n
service	internal/server	memory system statistics endpoint	done	yes	unknown	unit	no	n
service	internal/server	WebSocket real-time communication	done	partial	unknown	unit	no	n
service	internal/session	session lifecycle (create/start/complete/pause)	done	yes	yes	unit	no	n
service	internal/session	session modes (planning/building/	done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Notes
		testing/ refactoring)						
service	internal/ session	focus chain integration	done	yes	yes	unit	no	n
service	internal/ session	context + metadata tracking	done	yes	yes	unit	no	n
service	internal/ session	session querying + filtering	done	yes	yes	unit	no	n
infrastructure	internal/ substrate	substrate abstraction layer	partial	partial	unknown	unit	no	n
service	internal/task	task creation + assignment	done	yes	yes	unit	no	n
service	internal/task	task status lifecycle	done	yes	yes	unit	no	n
service	internal/task	priority-based task queue	done	yes	yes	unit	no	n
service	internal/task	checkpoint creation + recovery	done	yes	yes	unit	no	n
service	internal/task	dependency + circular- dependency validation	done	yes	yes	unit	no	n
service	internal/task	dependent task tracking	done	yes	yes	unit	no	n
service	internal/task	redis caching for tasks	done	yes	yes	unit	no	n
service	internal/ telemetry	agent instrumentation (prompt/ response)	done	partial	yes	unit	no	n
service	internal/ telemetry	LLM instrumentation (calls/latency/ tokens)	done	partial	yes	unit	no	n
service	internal/ telemetry	tool instrumentation (execution/ errors)	done	partial	yes	unit	no	n
service	internal/ telemetry	OpenTelemetry provider integration	done	partial	yes	unit	no	n
service			done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
	internal/ template	template creation + registration						
service	internal/ template	variable substitution + validation	done	yes	yes	unit	no	n
service	internal/ template	built-in template library	done	yes	yes	unit	no	n
service	internal/theme	theme detection (OS/ system prefs)	done	yes	yes	unit	no	n
service	internal/theme	theme loading + customization	done	yes	yes	unit	no	n
service	internal/theme	built-in theme collection	done	yes	yes	unit	no	n
service	internal/tools	file read/write operations	done	yes	yes	unit	no	n
service	internal/tools	file editing with string replacement	done	yes	yes	unit	no	n
service	internal/tools	file globbing (pattern matching)	done	yes	yes	unit	no	n
service	internal/tools	file content search (grep)	done	yes	yes	unit	no	n
service	internal/tools	shell command execution	done	yes	yes	unit	no	n
service	internal/tools	background shell execution	done	yes	yes	unit	no	n
service	internal/tools	web page fetching + parsing	done	yes	unknown	unit	no	n
service	internal/tools	web search integration	done	yes	unknown	unit	no	n
service	internal/tools	browser automation (launch/ navigate/ screenshot)	done	yes	unknown	unit	no	n
service	internal/tools	codebase mapping + symbol definitions	done	yes	yes	unit	no	n
service	internal/tools		done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	✓
		multi-file transactional editing						
service	internal/tools	user confirmation prompts	done	yes	yes	unit	no	n
service	internal/tools	notebook read/edit operations	done	yes	yes	unit	no	n
infrastructure	internal/tools	tool registry + execution	done	yes	yes	unit	no	n
service	internal/verifier	LLMsVerifier HTTP client	done	yes	unknown	unit,integ	no	n
service	internal/verifier	model metadata adapter	done	yes	unknown	unit	no	n
infrastructure	internal/verifier	two-tier cache (LRU + Redis)	done	partial	unknown	unit	no	n
infrastructure	internal/verifier	health monitoring + circuit breaker	done	partial	unknown	unit	no	n
infrastructure	internal/verifier	background poller (real-time updates)	done	partial	unknown	unit	no	n
infrastructure	internal/verifier	event publishing to HelixCode bus	done	partial	unknown	unit	no	n
service	internal/version	version string retrieval	done	yes	yes	unit	no	n
service	internal/version	build metadata exposure	done	yes	yes	unit	no	n
service	internal/voice	audio capture (arecord/sox)	partial	partial	unknown	unit	no	n
service	internal/voice	Whisper API transcription	partial	partial	unknown	unit	no	n
service	internal/voice	local whisper.cpp fallback	partial	no	unknown	unit	no	n
service	internal/worker	worker registration + management	done	yes	yes	unit	no	n
service	internal/worker	SSH-based remote execution	done	yes	yes	unit	no	n
service	internal/worker	health monitoring + metrics	done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
service	internal/worker	capability auto-detection	done	yes	yes	unit	no	no
service	internal/worker	worker isolation + sandboxing	done	partial	unknown	unit	no	no
service	internal/worker	consensus protocol (leader election)	done	partial	unknown	unit	no	no
service	internal/worker	host-key verification + SSH security	done	yes	yes	unit	no	no
service	internal/workflow	planning workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	building workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	testing workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	refactoring workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	DAG-based step orchestration	done	yes	yes	unit	no	no
service	internal/workflow	LLM provider integration for workflows	done	partial	unknown	unit	no	no
infrastructure	internal/workspace	container-based workspace management	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	Docker/Podman container orchestration	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	project directory mounting + isolation	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	auto-cleanup TTL enforcement	partial	no	unknown	unit	no	no
infrastructure	internal/mocks	unit-test mock fixtures (test-	done	yes	n/a	none	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	
		only, not a user feature)						
infrastructure	internal/pproftutil	pprof profiling helper (dev/test support)	done	partial	unknown	none	no	n
infrastructure	internal/testutil	shared test utilities (test-only, not a user feature)	done	yes	n/a	none	no	n

Coverage notes

- **Partially assessed (need deeper inventory):** clientcore, agentbridge, checkpoint, ensembleui, substrate, workspace, voice, roocode, telemetry, verifier (poller/event-bus wiring), worker (isolation/consensus), and server real-use (declared endpoints, no integration evidence inspected). Their core types compile and have unit tests, but wiring into a shipped flow and genuine end-user reachability could not be fully confirmed from static inspection alone — flagged partial/unknown honestly rather than green.
- mocks, pprofutil, testutil are dev/test-support packages, not user features — listed for completeness, marked n/a.
- Every Real-use=unknown and every working-untaped row is a candidate for a recorded scenario; none is video-confirmed yet (📹 no throughout).

233 features inventoried across 72 packages.

Sources verified 2026-06-22: helix_code/internal/*

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced repo trees, following the docs/ARCHITECTURE.md precedent — no external service documented). Cross-referenced against the live tree on 2026-06-22: - **helix_code/internal/* = 72 packages — CONFIRMED** (ls -d helix_code/internal/*/ = 72), matching the “across 72 packages” rollup. The per-row Dev/Wired/Tests assessments are structural evidence (impl reality + _test.go presence) about HelixCode’s own code. - No external-service version claims in this doc → no §11.4.99 staleness check applies; nothing to contradict against an upstream source.